ABBREVIATIONS

A/C	ABOVE CEILING
AC	AIR CONDITIONING
AD	ACCESS DOOR
ADJ	ADJUSTABLE
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AUTO	AUTOMATIC
B/G	BELOW GRADE
B/S	BELOW SLAB
BAL	BALANCING
BCO	BASE CLEANOUT
BDD	BACKDRAFT DAMPER
BFLY	BUTTERFLY
BHP	BRAKE HORSEPOWER
BOD	BASIS OF DESIGN
BOD	BOTTOM OF DUCT
CD	CEILING DIFFUSER
CFM	CUBIC FEET PER MINUTE
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
CO	CLEANOUT
COND	CONDENSATE
CSR	CURVED SUPPLY REGISTER
CU	CONDENSING UNIT
CW	COLD WATER (DOMESTIC)
CWR	CONDENSER WATER RETURN
CWS	CONDENSER WATER SUPPLY
dB	DECIBELS
DB	DRY BULB
DN	DOWN
DR	DRAIN
DWG	DRAWING
EA	EACH
EAT	ENTERING AIR TEMP.
ECC	ECCENTRIC
EF	EXHAUST FAN
EFF	EFFICIENCY
EG	EXHAUST GRILLE
ER	EXHAUST REGISTER
ESP	EXTERNAL STATIC PRESSURE
EWT	ENTERING WATER TEMP.
EX	EXISTING
EXH	EXHAUST
F	FAHRENHEIT
FCO	FLOOR CLEANOUT
FCU	FAN COIL UNIT
FD	FIRE DAMPER
FLR	FLOOR
FOB	FLAT ON BOTTOM
FOR	FUEL OIL RETURN
FOS	FUEL OIL SUPPLY
FOT	FLAT ON TOP
FPB	FAN POWERED BOX
FPM	FEET PER MINUTE
FPS	FEET PER SECOND
FSD	FIRE/SMOKE DAMPER
FT	FEET
G	GATE
GA	GAUGE
GPM	GALLONS PER MINUTE
HD	HUB DRAIN
HP	HORSEPOWER
HTG	HEATING
HW	HOT WATER (DOMESTIC)
HWR	HEATING HOT WATER RETURN
HWS	HEATING HOT WATER SUPPLY
HWRR	HOT WATER REVERSE RETURN
ID	INSIDE DIMENSION
IN	INCHES
KW	KILOWATTS
LAT	LEAVING AIR TEMPERATURE
LB	POUNDS
LSD	LINEAR SLOT DIFFUSER
LWR	LOOP WATER RETURN
LWS	LOOP WATER SUPPLY
MAX	MAXIMUM
MD	MANUAL DAMPER
MIN	MINIMUM
MOD	MOTOR OPERATED DAMPER
MFR	MANUFACTURER
NC	NORMALLY CLOSED
NG	NATURAL GAS
NO	NORMALLY OPEN
NOM	NOMINAL
OA	OUTSIDE AIR
OBD	OPPOSED BLADE DAMPER
OD	OUTSIDE DIMENSION
PIU	POWERED INDUCTION UNIT
PSI	POUNDS PER SQUARE INCH
RA	RETURN AIR
RAD	RADIUS
RAG	RETURN AIR GRILLE
RAR	RETURN AIR REGISTER
RED	REDUCER
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
RTU	ROOFTOP UNIT
SA	SUPPLY AIR
SAN	SANITARY
SD	SMOKE DAMPER
SD	SLOT DIFFUSER
SEN	SENSIBLE
SP	STATIC PRESSURE
SQ	SQUARE
SR	SUPPLY REGISTER
SS	SPLIT SYSTEM
ST	STORM
TEMP	TEMPERATURE
TG	TRANSFER GRILLE
TYP	TYPICAL
UON	UNLESS OTHERWISE NOTED
V	VENT
VA	VALVE
VAV	VARIABLE AIR VOLUME
VTR	VENT THRU ROOF
wb	WET BULB
WC	WATER COLUMN
WHA	WATER HAMMER ARRESTOR
WT	WEIGHT
W	WASTE

APPLICABLE CODES

- International Building Code, 2024 Edition, with Georgia Amendments (2026)
- International Fire Code, 2024 Edition
- International Mechanical Code, 2024 Edition, with Georgia Amendments (2026)
- International Fuel Gas Code, 2024 Edition, with Georgia Amendments (2026)
- International Energy Conservation Code, 2015 Edition, with Georgia Supplements and Amendments

MECHANICAL SHEET INDEX:

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M-503	MECHANICAL CONTROLS DETAILS

SPECIFICATIONS

PART 1 - GENERAL

1. SUBMITTALS
   A. PRODUCT DATA: FOR EACH TYPE OF PRODUCT INDICATED.

2. WARRANTY WARRANT NEW AND MODIFIED DUCTWORK FOR PERIOD OF ONE YEAR FROM DATE OF FINAL ACCEPTANCE OF JOB, AGAINST NOISE AND VIBRATION UNDER FULL RANGE OF OPERATING CONDITIONS.

PART 2 - PRODUCTS

1. METAL DUCTS
   A. CONTINUOUS HOT-DIP MILL GALVANIZED. MINIMUM COATING OF G60, LOCK-FORMING QUALITY STEEL SHEETS IN ACCORDANCE WITH ASTM A653.
   B. GAUGES: SHEET METAL GAUGE AS SPECIFIED IN SMACNA - HVAC DUCT CONSTRUCTION STANDARDS BUT NOT LESS THAN THE FOLLOWING:
      - RECTANGULAR DUCT: 26 GAUGE FOR ALL SIZES.
      - ROUND DUCT: PRIME GRADE STEEL SHEET.
        a. 14 INCH DIAMETER AND SMALLER: 26 GAUGE.
        b. 15 INCH DIAMETER THRU 26 INCH DIAMETER: 24 GAUGE.
        c. ROUND DUCT SHALL BE SPIRAL SEAM WHERE EXPOSED AND NOT INSULATED. IN ALL OTHER AREAS SPIRAL OR LONGITUDINAL SHALL BE USED.

2. TURNING VANES
   A. MANUFACTURED TURNING VANES FOR METAL DUCTS: CURVED BLADES OF GALVANIZED SHEET STEEL; SUPPORT WITH BARS PERPENDICULAR TO BLADES SET; SET INTO VANE RUNNERS SUITABLE FOR DUCT MOUNTING.
   B. GENERAL REQUIREMENTS: COMPLY WITH SMACNA "HVAC DUCT CONSTRUCTION STANDARDS - METAL AND FLEXIBLE"; FIGURES 4-3 "VANES AND VANE RUNNERS," 4-4, "VANE SUPPORT IN ELBOWS."
   C. VANE CONSTRUCTION: SINGLE WALL

3. SEALANT AND GASKET

   A. GENERAL SEALANT AND GASKET REQUIREMENTS: SURFACE-BURNING CHARACTERISTICS FOR SEALANTS AND GASKETS SHALL BE A MAXIMUM FLAME-SPREAD INDEX OF 25 AND A MAXIMUM SMOKE- DEVELOPED INDEX OF 50 WHEN TESTED ACCORDING TO UL 723; CERTIFIED BY AN NRTL.

4. HANGERS AND SUPPORTS

   A. HANGER RODS: CADMIUM-PLATED STEEL RODS AND NUTS.
   B. STRAP AND ROD SIZES: COMPLY WITH SMACNA'S "HVAC DUCT CONSTRUCTION STANDARDS - METAL AND FLEXIBLE" TABLE 5-1." RECTANGULAR DUCT HANGERS MINIM SIZE." AND TABLE 5-2 "MINIMUM HANGER SIZES FOR ROUND DUCT."
   C. STEEL CABLES FOR GALVANIZED STEEL DUCTS: GALVANIZED STEEL COMPLYING WITH ASTM A 603.
   D. STEEL CABLE END CONNECTIONS: CADMIUM PLATED STEEL ASSEMBLIES WITH BRACKETS, SWIVEL, AND BOLTS DESIGNED FOR DUCT HANGER SERVICE; WITH AN AUTOMATIC-LOCKING AND CLAMPING DEVICE.
   E. DUCT ATTACHMENTS: SHEET METAL SCREWS, BLIND RIVETS, OR SELF-TAPPING METAL SCREWS; COMPATIBLE WITH DUCT MATERIALS.

5. MECHANICAL EQUIPMENT

   A. PROVIDE EQUIPMENT INDICATED ON DRAWINGS.
   B. PRODUCTS REQUIRING ELECTRICAL CONNECTION: LISTED AND CLASSIFIED BY UNDERWRITERS LABORATORIES, INCORPORATED, AND ACCEPTABLE TO THE AUTHORITY HAVING JURISDICTION AS SUITABLE FOR THE PURPOSE SPECIFIED AND INDICATED.
   C. OPERATION AND MAINTENANCE DATA: INCLUDE MANUFACTURER'S DESCRIPTIVE LITERATURE. OPERATING INSTRUCTIONS, MAINTENANCE AND REPAIR DATA, AND PARTS LISTING FOR MECHANICAL EQUIPMENT.

6. CONTROLS

   A. PROVIDE COMPLETE CONTROLS SYSTEM INCLUDING STEP-DOWN TRANSFORMERS FOR 24 VAC CONTROL, RELAYS, RELAY BOARDS, RELAY BASES, PRINTED CIRCUIT BOARDS, THERMOSTATS, TIMERS, SENSORS, INDICATORS, ACTUATORS, AND OTHER NECESSARY DEVICES REQUIRED FOR COMPLETE INSTALLATION INCLUDING ENCLOSURES FOR EQUIPMENT. LOCATIONS AND TYPES OF THERMOSTATS, TIMERS, AND OTHER CONTROLS EQUIPMENT ARE INDICATED ON DRAWINGS.
   B. POWER SUPPLIES: TRANSFORMERS WITH CLASS 2 CURRENT-LIMITING TYPE OR OVERCURRENT PROTECTION; LIMIT CONNECTED LOADS TO 80 PERCENT OF RATED CAPACITY. DC POWER SUPPLY SHALL MATCH OUTPUT CURRENT AND VOLTAGE REQUIREMENTS AND BE FULL WAVE RECTIFIER TYPE.
   C. DEVICES
      - TEMPERATURE SENSORS
        a. ACCURACY: PLUS OR MINUS 0.36 DEG F OR 0.2 PERCENT AT CALIBRATION POINT.
        b. WIRE: TWISTED SHIELDED PAIR - CABLE.
        c. OUTSIDE AIR SENSORS: WATERTIGHT INLET FITTING, SHIELDED FROM DIRECT SUNLIGHT.
      - PROGRAMMABLE THERMOSTAT
        a. ELECTRIC, SOLID STATE, MICROCOMPUTER-BASED ROOM THERMOSTAT WITH REMOTE SENSOR.
        b. AUTOMATIC SWITCHING FROM HEATING TO COOLING.
      - LOW VOLTAGE ON / OFF THERMOSTATS
        a. MC TO INSTALL A TEMPORARY DIGITAL THERMOSTAT FOR A START-UP. FINAL THERMOSTAT TO BE KMC FLEXSTAT PROVIDED AND INSTALLED BY CONTROLS VENDOR.
   D. LOW VOLTAGE CONTROL WIRING
      - JACKETED, INDIVIDUALLY SHIELDED TWISTED PAIRS OF STRANDED INSULATED TINNED COPPER CONDUCTORS.
      - STRANDED TINNED COPPER DRAIN WIRE.
      - PLENUM-RATED JACKET FOR PLENUM APPLICATIONS.
      - CONDUCTOR SIZES BASED ON 10% MAXIMUM VOLTAGE DROP.
      - MINIMUM WIRE SIZE: 22 AWG, MAXIMUM WIRE SIZE: 14 GAUGE.

PART 3 - EXECUTION

1. GENERAL

   A. UNLESS OTHERWISE DICTATED BY APPLICABLE CODES OR THE AUTHORITY HAVING JURISDICTION, ALL MATERIALS WITHIN RETURN AIR PLENUM MUST BE NONCOMBUSTIBLE AND/OR HAVE A FLAME SPREAD INDEX OF NOT MORE THAN 25 AND A SMOKE-DEVELOPED INDEX NOT MORE THAN 50 IN ACCORDANCE WITH ASTM E 84 TESTING.
   B. PROTECT WORK, EQUIPMENT AND MATERIAL TO PREVENT OBSTRUCTION, DAMAGE OR BREAKAGE. CLOSE PIPE OPENINGS WITH CAPS OR PLUGS DURING INSTALLATION. COVER AND PROTECT EQUIPMENT AGAINST DIRT, WATER, CHEMICAL OR MECHANICAL INJURY. AT THE COMPLETION OF WORK, THOROUGHLY CLEAN ALL EQUIPMENT AND DELIVER THE ENTIRE SYSTEM IN AN UNBLEMISHED CONDITION.
   C. MAKE CHANGES IN PULLEYS, BELTS, DUCTWORK, AND DAMPERS AS REQUIRED FOR CORRECT BALANCE AS RECOMMENDED BY AIR BALANCE AGENCY.
   D. PROVIDE PENETRATION FIRESTOPPING THAT IS PRODUCED AND INSTALLED TO MAINTAIN ORIGINAL FIRE-RESISTANCE RATING OF CONSTRUCTION PENETRATED. PENETRATION FIRESTOPPING SYSTEMS SHALL BE COMPATIBLE WITH ONE ANOTHER, WITH THE SUBSTRATES FORMING OPENING, AND WITH PENETRATING ITEMS IF ANY.
      - COORDINATE SIZING OF SLEEVES, OPENINGS CORE-DRILLED HOLES, OR CUT OPENINGS TO ACCOMMODATE PENETRATION FIRESTOPPING.
      - INSTALL PENETRATION FIRESTOPPING TO COMPLY WITH MANUFACTURER'S WRITTEN INSTALLATION INSTRUCTIONS AND PUBLISHED DRAWINGS FOR PRODUCTS AND APPLICATIONS INDICATED.

2. DUCT INSTALLATION
   A. INSTALL DUCTS ACCORDING TO SMACNA STANDARDS.
   B. PROTECT DUCT INTERIORS FROM MOISTURE, CONSTRUCTION DEBRIS, AND OTHER FOREIGN MATERIALS.
   C. FABRICATE, ERECT, AND INSTALL DUCTWORK FOR HEATING, VENTILATING, AND AIR CONDITIONING PER SMACNA STANDARDS AND THE REQUIREMENTS OF THE AUTHORITY HAVING JURISDICTION.
   D. MAKE JOINTS AIR TIGHT ON ALL DUCTWORK. SEAL EXHAUST DUCTWORK WITH EXTERIOR SEAL AS REQUIRED TO ASSURE POSITIVE SEAL. COMPLY WITH SMACNA DUCT CLASS A.
   E. INSTALL FACTORY FABRICATED FITTINGS FOR CHANGES IN DIRECTION, SIZE, AND SHAPE AND FOR BRANCH CONNECTIONS.

3. HANGERS AND SUPPORT INSTALLATION
   A. COMPLY WITH SMACNA STANDARDS.
   B. INSTALL HANGERS AND SUPPORTS WITHIN 24" OF EACH ELBOW AND WITH 48" OF EACH BRANCH INTERSECTION.
   C. DO NOT USE FASTENERS THAT PENETRATE ROOF DECK.

4. MECHANICAL EQUIPMENT
   A. INSTALL PER MANUFACTURER'S RECOMMENDATIONS.

5. CONTROLS
   A. COORDINATE LOCATION OF THERMOSTATS, HUMIDISTATS, AND OTHER EXPOSED CONTROL SENSORS WITH PLANS AND ROOM DETAILS BEFORE INSTALLATION.
   B. CONNECT AND CONFIGURE EQUIPMENT AND SOFTWARE TO ACHIEVE SEQUENCE OF OPERATIONS.
   C. LOW VOLTAGE CONTROL WIRING.
      - COMPLY WITH NECA 1.
      - COMPLY WITH TIA/EIA-568-B.1.
      - BUNDLE AND HARNESS MULTICONDUCTOR INSTRUMENT CABLE IN PLACE OF SINGLE CABLES WHERE SEVERAL CABLES FOLLOW A COMMON PATH.
      - NUMBER-CODE OR COLOR-CODE CONDUCTORS FOR FUTURE IDENTIFICATION AND SERVICE OF CONTROL SYSTEM, EXCEPT LOCAL INDIVIDUAL ROOM CONTROL CABLES.
      - CONCEAL CABLE, EXCEPT IN MECHANICAL ROOMS AND AREAS WHERE OTHER CONDUIT AND PIPING ARE EXPOSED.
      - INSTALL PLENUM-RATED CABLE IN ENVIRONMENTAL AIR SPACES INCLUDING PLENUM CEILINGS.
      - WIRING WITHIN ENCLOSURES
        a. BUNDLE, LACE, AND TRAIN CONDUCTORS TO TERMINAL POINTS WITH NO EXCESS.
        b. INSTALL CONDUCTORS PARALLEL, WITH OR AT RIGHT ANGLES TO SIDES AND BACK OF ENCLOSURE.
   D. AFTER ELECTRICAL CIRCUITRY HAS BEEN ENERGIZED, START UNITS TO CONFIRM PROPER UNIT OPERATION. REMOVE AND REPLACE MALFUNCTIONING UNITS AND RETEST.
   E. TEST EACH SYSTEM FOR COMPLIANCE WITH SEQUENCE OF OPERATION.
   F. OCCUPANCY ADJUSTMENTS, WHEN REQUESTED WITHIN 12 MONTHS OF DATE OF SUBSTANTIAL COMPLETION, PROVIDE ON-SITE ASSISTANCE IN ADJUSTING SYSTEM TO SUIT ACTUAL OCCUPIED CONDITIONS. PROVIDE UP TO THREE VISITS TO PROJECT DURING OTHER THAN NORMAL OCCUPANCY HOURS FOR THIS PURPOSE.

6. EQUIPMENT LABEL INSTALLATION

   A. INSTALL OR PERMANENTLY FASTEN LABELS ON EACH MAJOR ITEM OF MECHANICAL EQUIPMENT.
   B. LABEL ACCESS DOORS TO INDICATE THE PURPOSE OF ACCESS DOOR.
   C. LOCATE EQUIPMENT LABELS WHERE ACCESSIBLE AND VISIBLE.
   D. WHERE EQUIPMENT, SUCH AS A VAV BOX, IS CONCEALED BY ACCESSIBLE DROP OR LAY-IN CEILINGS OR OTHER ACCESSIBLE BARRIER, ATTACH LABEL TO CEILING GRID OR A VISIBLE LOCATION TO IDENTIFY LOCATION OF HIDDEN EQUIPMENT.

7. TESTING, ADJUSTING, AND BALANCING

   A. PERFORM TESTING AND BALANCING PROCEDURES ON EACH SYSTEM ACCORDING TO THIS SECTION AND THE PROCEDURES CONTAINED IN ONE OF THE FOLLOWING:
      - AABC'S "NATIONAL STANDARDS FOR TOTAL SYSTEM BALANCE"
      - NEBB'S "PROCEDURAL STANDARDS FOR TESTING, ADJUSTING, AND BALANCING OF ENVIRONMENTAL SYSTEMS."
   B. COMPLY WITH REQUIREMENTS IN ASHRAE 62.1 SECTION 7.2.2 - "AIR BALANCING."
   C. AIRFLOW VALUES ARE TO BE ADJUSTED TO PLUS OR MINUS 5% OF DESIGN FLOW.
   D. SET DAMPERS TO FAIL SAFE POSITION BEFORE TESTING, ADJUSTING, AND BALANCING.
   E. MARK EQUIPMENT AND BALANCING DEVICES, INCLUDING DAMPER-CONTROL POSITIONS, VALVE POSITION INDICATORS, FAN-SPEED-CONTROL LEVERS, AND SIMILAR CONTROLS AND DEVICES, WITH PAINT OR OTHER SUITABLE, PERMANENT IDENTIFICATION MATERIAL TO SHOW FINAL SETTINGS.
   F. ADJUST FANS TO DELIVER DESIGN AIRFLOW WITHIN THE DESIGN FAN SPEED LISTED BY FAN MANUFACTURER. ANY ADJUSTMENTS MUST BE COMMUNICATED WITH CONTROLS VENDOR WHILE T&B IS ON SITE TO CONFIRM ON BMS AND DOCUMENT THE CHANGE ON THE UI.
   G. AFTER TESTING AND BALANCING IS COMPLETE, OPERATE EACH SYSTEM AND RANDOMLY CHECK MEASUREMENTS TO VERIFY THAT THE SYSTEMS ARE OPERATING IN ACCORDING TO THE FINAL TEST AND BALANCE READINGS.

8. OPERATION PRIOR TO ACCEPTANCE: CONTRACTOR MAY OPERATE ANY EQUIPMENT PROVIDED THAT THE OPERATION IS SUPERVISED AND THE CONTRACTOR RETAINS FULL RESPONSIBILITY FOR PROPERLY MAINTAINING EQUIPMENT AND THE FULL MANUFACTURER'S WARRANTY REMAINS UNAFFECTED FROM THE TIME OF OWNER'S POSSESSION.

GENERAL NOTES

A. CONTRACTOR MAY OPERATE ANY EQUIPMENT PROVIDED THAT THE OPERATION IS SUPERVISED AND THE CONTRACTOR RETAINS FULL RESPONSIBILITY FOR PROPERLY MAINTAINING EQUIPMENT AND THE FULL MANUFACTURER'S WARRANTY REMAINS UNAFFECTED FROM THE TIME OF OWNER'S POSSESSION.

B. CONTRACTOR SHALL FURNISH A WRITTEN GUARANTEE COVERING ALL WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE YEAR FROM DATE OF ACCEPTANCE.

C. CONTRACTOR SHALL PROVIDE POSITIVE MEANS FOR BALANCING EACH INDIVIDUAL AIR CONDITIONING SUPPLY AIR OUTLET, AS PER SCHEDULES & DRAWINGS, SYSTEMS SHALL BE BALANCED AGAINST THE ACTUAL INSTALLED STATIC PRESSURE.

D. BEFORE SUBMITTING A FINAL PROPOSAL, THE CONTRACTOR SHALL EXAMINE THE SITE OF THE PROPOSED WORK TO DETERMINE THE EXISTING CONDITIONS THAT MAY AFFECT HIS WORK AS HE WILL BE RESPONSIBLE FOR ANY ASSUMPTIONS MADE BY HIM IN REGARDS HERETO.

E. ANY APPARATUS, APPLIANCE, MATERIAL, WORK OR INCIDENTAL ACCESSORIES OR MINOR DETAILS NOT SHOWN BUT NECESSARY TO MAKE THE WORK COMPLETE AND PERFECT IN ALL RESPECTS AND READY FOR OPERATION, EVEN IF NOT PARTICULARLY SPECIFIED SHALL BE PROVIDED BY THE CONTRACTOR WITHOUT ADDITIONAL EXPENSE TO THE OWNER.

F. CONTRACTOR SHALL ASSUME RESPONSIBILITY FOR ALL COSTS INCURRED RESULTING FROM SUBSTITUTION OF EQUIPMENT AS WELL AS THE PERFORMANCE AND SPACE REQUIREMENTS OF SUCH EQUIPMENT.

G. ALL TESTS SHALL BE PERFORMED AS REQUIRED DURING THE DIFFERENT STAGES OF WORK AND A FINAL 24 HOUR MINIMUM RUNNING TEST SHALL BE DONE AFTER ALL OTHER TESTS AND BALANCING OPERATIONS HAVE BEEN DONE.

H. TEST AND BALANCE OF ALL EQUIPMENT SHALL BE PERFORMED BY AN INDEPENDENT TEST AND BALANCE COMPANY WITH A MINIMUM OF 5 YEARS EXPERIENCE IN PROJECTS OF THIS SIZE OR LARGER. THREE COPIES OF THE TEST AND BALANCE REPORT SHALL BE SUBMITTED TO THE ENGINEER FOR REVIEW AND APPROVAL.

I. CONTRACT DOCUMENTS FOR THE WORK ARE BASED ON AS-BUILT DRAWINGS PROVIDED BY OWNER. ANY DISCREPANCIES BETWEEN THE WORK DEPICTED IN THE DOCUMENTS AND THE ACTUAL CONDITIONS ENCOUNTERED SHALL BE BROUGHT PROMPTLY TO THE EOR FOR REVIEW AND CLARIFICATION.

J. ANY ADDITIONAL DUCTWORK REQUIRED TO MAKE FINAL EXHAUST DUCTWORK CONNECTIONS SHALL BE METAL PER FMC 501.4. CONSTRUCTION SHALL COMPLY WITH FMC 603. ANY DEFICIENCIES IN EXISTING DUCTWORK CONSTRUCTION SHALL BE REMOVED AND MADE COMPLIANT WITH THE ABOVE LISTED CODE SECTIONS.

K. ALL INDICATED AIR FLOWS ARE THE MINIMUM REQUIRED FOR THE ZONE. IF MORE AIR FLOW IS PROVIDED DISTRIBUTE THE ACCESS AIR PROPORTIONALLY, BY AREA, TO EACH ZONE SUPPLIED BY THE SYSTEM.

L. ALL RUNOUTS TO AIR TERMINALS TO BE SAME SIZE AS AIR TERMINAL CONNECTION UNLESS OTHERWISE NOTED.

M. PROVIDE VOLUME BALANCING DAMPERS FOR ALL DUCT BRANCHES CONNECTED TO AIR TERMINALS. ALL DUCTWORK TO BE SEALED SMACNA CLASS A.

N. T-STATS AND SENSORS TO BE MOUNTED NO MORE THAN 48" ABOVE FINISHED FLOOR, UNLESS OTHERWISE NOTED. PROVIDE 55° F DISCHARGE AIR TEMPERATURE FROM AIR HANDLING UNIT. ALL REQUIRED THERMOSTATS SHALL BE LOCATED AS SHOWN ON DRAWINGS OR WITHIN THE ZONE OF THE UNIT THEY ARE CONTROLLING, UNLESS OTHERWISE NOTED.

O. SUPPORT SUSPENDED MECHANICAL EQUIPMENT WITH ALL-THREAD RODS AND VIBRATION ISOLATORS.

P. PROVIDE GENERAL CONTRACTOR WITH FINAL SIZES AND LOCATIONS OF ALL WALL, FLOOR, AND ROOF PENETRATIONS TO COORDINATE ANY REQUIRED STRUCTURAL FRAMING MEMBERS.

Q. ALL PENETRATIONS OF DUCTWORK, PIPING CONDUITS, AND VENTS THROUGH FIRE-RATED AND SMOKE-RATED BARRIERS SHALL HAVE FIRESTOP AND/OR SMOKE STOP PROTECTION IN ACCORDANCE WITH THE MINIMUM BUILDING CODE.

R. IF AC UNITS MUST BE USED DURING CONSTRUCTION, THEY SHALL HAVE FILTRATION MEDIA WITH A MERV 8 AT EACH RETURN AIR GRILLE. THESE MUST BE REPLACED PRIOR TO OCCUPANCY.

S. HOLD ALL PIPING AND DUCTWORK TIGHT TO STRUCTURE IN ALL AREAS. NOTE THERE ARE HIGH CEILINGS IN MANY AREAS WHICH WILL REQUIRE CLOSE COORDINATION WITH OTHER TRADES DURING SHOP DRAWING PREPARATION AND DURING CONSTRUCTION.

T. SMOKE DETECTORS AND REMOTE ANNUNCIATOR ARE SUPPLIED BY MECHANICAL CONTRACTOR. SMOKE DETECTORS SHALL BE INSTALLED AND INTERLOCKED FOR SHUTDOWN IN ACCORDANCE WITH DIV. 23. SEE SPECIFICATIONS.

U. THREE SETS OF FILTERS ARE TO BE PROVIDED BY CONTRACTOR FOR EACH AC UNIT. ONE SET IS FOR USE DURING CONSTRUCTION, ONE SET TO BE INSTALLED PRIOR TO TEST AND BALANCE, AND A FINAL SET TO BE INSTALLED JUST PRIOR TO OCCUPANCY.

V. ALL MECHANICAL EQUIPMENT APPROACHING END OF USEFUL LIFE (15 YEARS) WILL BE REPLACED WITH LIKE SIZE EQUIPMENT. EFFICIENCIES SHALL BE COMPLIANT WITH LOCAL CODES.

W. CONTRACTOR SHALL BE RESPONSIBLE FOR INSTALLING ALL NECESSARY ACCESS PANELS TO PROVIDE ACCESS TO CONCEALED EQUIPMENT AND BALANCE DAMPERS, REGARDLESS OF WHETHER SAID ACCESS PANELS ARE SHOWN ON THE PLANS.

X. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE HVAC SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

Y. CERTAIN ITEMS SUCH AS RISES AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS.

Z. IN CORRIDORS WHERE CEILING SPEAKERS AND AIR DIFFUSERS ARE INDICATED BETWEEN THE SAME LIGHT FIXTURES, INSTALL BOTH DEVICES AT THE QUARTER POINTS BETWEEN THE SAME FIXTURE.

AA. ALL DUCTWORK SHALL CLEAR DOORS AND WINDOWS.

BB. UNLESS OTHERWISE NOTED, ALL DUCTWORK IS OVERHEAD, TIGHT TO THE UNDERSIDE OF THE STRUCTURE, WITH SPACE FOR INSULATION IF REQUIRED.

CC. RUNS OF FLEXIBLE DUCT SHALL NOT EXCEED 5 FEET (EDIT MAXIMUM LENGTH OF FLEXIBLE DUCT TO SUIT PROJECT; 5 FEET MAXIMUM RECOMMENDED LENGTH, 8 FEET MAXIMUM LENGTH).

DD. ALL DUCTWORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN DUCTS, INCLUDING DIVIDED DUCTS AND TRANSITIONS AROUND OBSTRUCTIONS, SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

EE. PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

FF. PROVIDE ACCESS DOORS IN DUCTWORK FOR OPERATION, ADJUSTMENT, AND MAINTENANCE OF ALL FANS, VALVES, AND MECHANICAL EQUIPMENT.

GG. ALL DUCTS SHALL BE GROUNDED ACROSS FLEXIBLE CONNECTIONS WITH FLEXIBLE COPPER GROUNDING STRAPS. GROUNDING STRAPS SHALL BE BOLTED OR SOLDERED TO BOTH THE EQUIPMENT AND THE DUCT.

HH. SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTRUCTIONS.

II. TERMINATE GAS VENTS FOR UNIT HEATERS, WATER HEATERS, HIGH PRESSURE PARTS WASHER, HIGH PRESSURE CLEANER, AND OTHER GAS APPLIANCES A MINIMUM OF 3'0" ABOVE ROOF WITH RAIN CAP (EDIT APPLIANCES AND HEIGHT ABOVE ROOF TO MEET CODE AND TO SUIT PROJECT REQUIREMENTS).

JJ. SEE SPECIFICATIONS FOR DUCTWORK GAUGES, BRACING, HANGERS, AND OTHER REQUIREMENTS.

KK. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. DETAILED DESCRIPTIONS ARE PROVIDED IN THE ARCHITECTURAL SPECIFICATIONS.

LL. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, AND DETAILS SHALL BE COORDINATED WITH GENERAL CONTRACTOR.

MM. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, MOUNTING, AND DETAILS SHALL BE COORDINATED WITH OTHER TRADES INVOLVED.

DISCLAIMER NOTE: NOT ALL SPECIFICATIONS OR NOTES ON THIS SHEET MAY APPLY TO THIS SPECIFIC PROJECT. REFER TO PROJECT MECHANICAL PLANS FOR SCOPE OF WORK APPLICABLE TO THIS PROJECT.

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:

FIFTH THIRD BANK

Project Number: 26032
Project Name:

FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:

MECHANICAL NOTES AND SPECIFICATIONS

Sheet Number:

M-001

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